Advanced Mathematics (Part 1) — Practice Problems

1. Limit Computation and Proofs

(1) Limits of Functions and Sequences

1. $\lim\limits_{n\to\infty}\left(\frac{1}{2}+\frac{3}{2^2}+\cdots+\frac{2n-1}{2^n}\right)$
2. $\lim\limits_{n\to\infty}\left(\sqrt{n+2}-2\sqrt{n+1}+\sqrt{n}\right)$
3. $\lim\limits_{x\to 1}\dfrac{\sqrt{x+8}-3}{\sqrt{x+3}-2}$
4. $\lim\limits_{x\to+\infty}\dfrac{(x-1)^2+1-3x}{x^2+2x^3}$
5. $\lim\limits_{x\to-\infty}\dfrac{\sqrt{x^2-1}}{x}$
6. $\lim\limits_{x\to 0}\dfrac{\sin x^2}{1-\cos x}$
7. $\lim\limits_{x\to 0}\dfrac{\sin x-\tan x}{x^3}$
8. $\lim\limits_{x\to 0}\left(1+2\tan^2 x\right)^{\cot^2 x}$
9. $\lim\limits_{x\to 0}\dfrac{\tan x-\sin x}{\sin^3 x}$
10. $\lim\limits_{x\to 0^+}\dfrac{\int_0^x t\sin t\,dt}{x^2\ln(1+x)}$
11. $\lim\limits_{x\to 0}\dfrac{\int_0^x \ln(1+t)\,dt}{x\left(e^x-1\right)}$
12. Find the following limits:

(1) $\lim\limits_{x\to\pi}\dfrac{1+\cos x}{\tan^2 x}$

(2) $\lim\limits_{x\to 0}\dfrac{\tan x-x}{x^2\tan x}$

(3) $\lim\limits_{x\to 1}x^{\frac{1}{1-x}}$

(4) $\lim\limits_{x\to 0^+}x^{\sin x}$

(5) $\lim\limits_{x\to 0}\dfrac{1-x^2-e^{-x^2}}{x\sin^3 2x}$

(6) $\lim\limits_{x\to 0}\left(\dfrac{1}{x}-\dfrac{\cos x}{\sin x}\right)\cdot\dfrac{1}{x}$

(2) Proof Problems

1. $f(x)$ is continuous on $[0,2]$ and $f(0)=f(2)$; prove that there exists at least one point $\xi\in[0,1]$ such that $f(\xi)=f(\xi+1)$
2. Prove that for $x\geq 1$, we have $2\arctan x+\arcsin\dfrac{2x}{1+x^2}=\pi$

3. Suppose the function $f(x)$ is continuous on the closed interval $[0,1]$ and differentiable on the open interval $(0,1)$, with $f(1)=0$; prove that in the open interval $(0,1)$ there exists at least one point ξ such that $f'(\xi)=-\dfrac{1}{\xi}f(\xi)$
4. Prove that $pb^{p-1}(a-b)\le a^p-b^p\le pa^{p-1}(a-b)$, where $0<b<a$ and $p>1$

2. Derivatives and Differentials

1. Find the equations of the tangent line and the normal line to the curve $\begin{cases}x=te^{-t}\\ y=e^{t}\end{cases}$ at the point where $t=0$.
2. Find the equation of the tangent line to the curve $y=\ln x$ that is parallel to the line $y=x$.
3. Let $y=e^{x}\ln x$; find y'.
4. Let $y=\left(\arctan\dfrac{x}{3}\right)^{2}$; find y'.
5. Let $y=\sin^{2}x\cdot\sin\left(x^{2}\right)$; find y'.
6. Let $y=\log_{2}\left(e^{x}-1\right)$; find y'.
7. Find the equation of the tangent line to the astroid $x^{\frac{2}{3}}+y^{\frac{2}{3}}=a^{\frac{2}{3}}\ (a>0)$ at the point $M\left(\dfrac{\sqrt{2}}{4}a,\dfrac{\sqrt{2}}{4}a\right)$.
8. Let the function be $y=\arctan\dfrac{1}{x}$; find dy and $dy\big|_{x=1}$.
9. Let $y=y(x)$ be determined by the equation $2^{xy}=x+y$; find $dy\big|_{x=0}$.
10. Given $y=\ln^{2}(1-x)$, find dy.
11. Given $y=\int_{0}^{x}\tan t^{2}\,dt$, find y'.

3. Indefinite and Definite Integrals

(1) Computing Indefinite Integrals

1. $\int\left(1-\sqrt[3]{x^{2}}\right)^{2}dx$
2. $\int\dfrac{1}{\cos^{2}x\sin^{2}x}\,dx$
3. $\int\dfrac{1}{4x^{2}+a^{2}}\,dx$
4. $\int x^{2}e^{-x^{3}}\,dx$
5. $\int\dfrac{e^{2x}}{1+e^{x}}\,dx$
6. $\int\sqrt{a^{2}-x^{2}}\,dx\ (a>0)$
7. $\int xe^{x}\,dx$
8. $\int x\ln x\,dx$

9. $\int \sec^3 x \, dx$

(2) Computing Definite Integrals

1. $\int_0^{\pi} \sqrt{1+\cos 2x} \, dx$
2. $\int_0^{2\pi} |\sin x| \, dx$
3. $\int_{-1}^{3} \dfrac{dt}{1+t^2}$
4. $\int_{\frac{1}{e}}^{e} |\ln x| \, dx$
5. $\int_0^{\pi^2} \sqrt{x} \cos \sqrt{x} \, dx$
6. $\int_0^{\frac{\pi}{2}} e^x \cos x \, dx$
7. $\int_0^2 x\sqrt{2x-x^2} \, dx$

4. Applications of Integration (Area, Volume, Arc Length)

1. Compute the area of the region enclosed by the parabola $y^2 = 2x$ and the line $y = x - 4$.
2. Compute the volume of the solid of revolution obtained by revolving the region enclosed by the ellipse $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$ one full turn about the x-axis.
3. A plane passes through the center of the base circle of a cylinder of base radius R and makes an angle α with the base plane; compute the volume of the solid cut from the cylinder by this plane.
4. Compute the area of the region enclosed by the cardioid $\rho = a(1+\cos\theta)$ $(a>0)$.
5. Find the arc length of the cardioid $\rho = a(1+\cos\theta)$ $(a>0)$.
6. The cycloid is given by $\begin{cases} x = a(t-\sin t) \\ y = a(1-\cos t) \end{cases}$ $(a>0)$ $(0 \le t \le 2\pi)$:

(1) Find the area of the region enclosed by one arch of the cycloid and the x-axis.

(2) Find the volumes of the solids of revolution obtained by revolving the region enclosed by one arch of the cycloid and the x-axis one full turn about the x-axis and the y-axis, respectively.

(3) Find the length of one arch of the cycloid.

5. Taylor's Formula and the Behavior of Functions

1. Find, for the function $f(x) = \sin x$, its nth-order Maclaurin formula with the Lagrange remainder.
2. Find, for the function $f(x) = e^x$, its nth-order Maclaurin formula with the Lagrange remainder.
3. Find the intervals of concavity and the inflection points of $y = \ln\left(1+x^2\right)$.

6. Power Series

1. Find the radius of convergence, open interval of convergence, and interval of convergence of the following power series:

(1) $\sum_{n=1}^{\infty} \frac{x^n}{2^n \cdot n}$

(2) $\sum_{n=1}^{\infty} \left(\sqrt{n+1} - \sqrt{n}\right)(x-2)^n$

(3) $\sum_{n=1}^{\infty} \frac{3^n + (-2)^n}{n} x^n$

1. Find the sum function of the power series $\sum_{n=1}^{\infty} \frac{x^{n+1}}{n}$, and find the sum of the series $\sum_{n=1}^{\infty} \frac{1}{n \cdot 2^n}$.
2. Find the sum function of the power series $\sum_{n=1}^{\infty} n x^n$, and use it to find the sum of the numerical series $\sum_{n=1}^{\infty} \frac{n}{2^n}$.
3. Find the sum function of $\sum_{n=1}^{\infty} \frac{x^n}{n(n+1)}$.